The Quantity–Risk Trade-off in Feedback Acquisition under Selective Labels

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Abstract. Lending decisions determine which repayment outcomes enter future training data. When feedback from a rejected region requires approving applicants, an acquisition score changes both the number of observed outcomes and the exposure used to obtain them. We define a maximum-feasible-count reference, K^*, and a same-count minimum-risk reference $M(k)$, then measure the quantity opportunity cost and predicted-risk premium of score-based selection. Random-order replay covers four credit datasets and five budget ratios; a registered five-policy block compares FAVE with bootstrap-committee information gain (IG). At $\rho = 0.1$, LRBE-IG acquires 31.3, 49.5, and 40.1 more labels than ScoreFirst-IG on HMEQ, LendingClub, and UCI. Relative to MinRisk, it uses 28,765.5 and 27,025.7 additional predicted-risk units on HMEQ and LendingClub, while Recall and Brier changes vary by dataset and direction. Same-state audits and a controlled retraining experiment connect these policy outcomes to immediate selection choices. The evidence supports evaluating acquisition scores by feedback quantity, predicted-risk use, composition, and downstream performance together.

Keywords: Selective labels · Feedback acquisition · Risk budgets · Credit prediction

1 Introduction

Credit decisions determine which repayment outcomes become available for later model training. Approved applicants generate observed outcomes, whereas rejected applicants remain selectively unlabeled. This dependence can make both evaluation and retraining reflect the policy that collected the data rather than the population of applicants [14,13]. In credit applications, obtaining feedback from a rejected region also requires accepting the associated predicted exposure. Feedback acquisition is therefore a resource-allocation problem: a selector chooses which outcomes to observe, how many to obtain, and how much predicted loss to spend.

Existing work provides tools for this setting. Reject inference corrects or augments samples affected by acceptance decisions [4,15]; selective-label methods study sequential decisions with one-sided feedback [21,10]; and exploration methods address self-confirming rejection, bounded debiasing, or controlled approvals [18,23,19]. Cost-sensitive active learning already treats sample costs as heterogeneous [8,20]. These strands establish the problem and several useful selection

scores, but final prediction metrics alone do not reveal the exchange among the number of labels obtained, the predicted risk spent, and the applicants selected.

We make that exchange explicit with lexicographic risk-budgeted feedback acquisition (LRBE). At a given policy state, let K_t^* be the largest number of candidate labels that fits the predicted-loss budget, and let $M_t(k)$ be the minimum predicted risk of any feasible set of size k. A score can then be evaluated against two concrete references: the maximum feasible count and the minimum risk required to retain that count. The quantity loss $D(S) = K_t^* - |S|$ records labels exchanged for score; the composition premium $P(S) = \widehat{R}(S) - M_t(|S|)$ records extra predicted risk at the selected count. This construction is a comparison benchmark, not a claim that sorting costs is a new optimization algorithm.

The empirical study answers three questions. First, how many feasible labels does a score-first policy give up? Dynamic replay on HMEQ, LendingClub, UCI, and PPDai shows that LRBE-IG obtains 31.3, 49.5, and 40.1 more labels than ScoreFirst-IG on HMEQ, LendingClub, and UCI at $\rho = 0.1$. Second, what risk premium accompanies a different composition at a similar cumulative count? Relative to MinRisk, LRBE-IG spends 28,765.5 additional predicted-risk units on HMEQ and 27,025.7 on LendingClub. Third, do these quantity and risk changes reach the prediction target? Recall and Brier differences vary in direction across datasets; UCI Brier error increases for the LRBE-IG versus ScoreFirst-IG contrast. Same-state audits confirm that IG changes selected sets, while the controlled experiment reproduces the immediate count–risk trade-off after one retraining step.

The paper makes three contributions. (1) It provides a measurable comparison benchmark based on the maximum feasible count K^* and the same-count minimum-risk profile $M(k)$, making the quantity opportunity cost and composition premium of an acquisition score visible. (2) It combines dynamic replay, same-state selection comparison, and controlled one-step retraining so that policy evolution, immediate selector behavior, and predictive outcomes are reported as distinct evidence blocks. (3) It gives an empirical finding across four credit datasets: score choice changes feedback quantity and predicted-risk use more consistently than it changes downstream prediction, whose direction is dataset-dependent. The resulting framework clarifies what a higher acquisition score buys and what it costs.

2 Related Work

Selective labels and controlled exploration. Selective-label research studies learning when decisions determine observed outcomes [14,13]. Expert consistency, fair decisions, and one-sided-feedback learning extend reasoning beyond existing acceptance patterns [6,12,10]. Reject inference and sample-selection correction address credit-specific observation bias [4,15]. Our replay uses observed cohort outcomes and reveals them under simulated approvals, so the estimand is policy behavior under replay.

Cost-sensitive and batch acquisition. Cost-sensitive active learning studies heterogeneous annotation costs and budgeted sample selection [8,20]. ConBatchBAL develops Bayesian batch acquisition under budget constraints, including dynamic thresholding and greedy selection based on uncertainty scores [16]. Batch-level uncertainty methods therefore provide a close reference for allocating a finite acquisition budget. Our comparison asks a narrower question in selective credit feedback: after the current maximum feasible count and the same-count minimum-risk profile are fixed, how much quantity and predicted-risk change is associated with a different composition score?

Exploration, safety, and evaluation. PLOT uses optimistic pseudo-labels to counter self-confirming rejection, while adaptive bounded exploration trades debiasing against accepting otherwise rejected individuals [18,23]. Partial-feedback exploration supports subgroup coverage and fairness objectives [11]; cost-efficient fairness auditing accounts for assessment, processing, and potential-default costs [5]; and controlled approvals in credit scoring examine retraining under selection bias [19]. These works supply complementary objectives and safeguards. LRBE uses a point predicted-loss allowance as a ledger, then reports the resulting quantity, composition, predicted-risk, and downstream changes together. The benchmark makes the cost of a score visible even when the score is useful for selecting a different set of applicants.

Resource constraints and predictive evaluation. Cost-sensitive active learning already accounts for heterogeneous query costs [8,20]. Contextual bandits with knapsack constraints and conservative exploration place learning within resource or baseline-performance restrictions [1,22]; safety-constrained approaches further consider uncertainty in feasibility [2,17]. LRBE uses a point predicted-loss constraint in place of these sequential guarantees. We evaluate secondary scores through the resulting data and predictions, comparing Recall and Brier while reporting predicted-budget compliance separately from realized defaulted exposure.

3 Risk-Budgeted Feedback Acquisition

3.1 Sequential observation and policy-specific budgets

At round t, policy m receives an incoming batch $\mathcal{U}_t$ and fits a default predictor on its visible labeled set $\mathcal{L}_t^{(m)}$. An outcome $y_i = 1$ denotes default. Ordinary approvals have predicted default probability below the fixed threshold $\tau = 0.5$, which serves as a common operating point for every policy in the replay. The remaining applicants form the exploration candidates

$$\mathcal{C}_t^{(m)} = \{i \in \mathcal{U}_t : \widehat{p}_{it}^{(m)} \geq \tau\}. \tag{1}$$

The policy selects exploratory approvals $\mathcal{S}_t^{(m)} \subseteq \mathcal{C}_t^{(m)}$. Outcomes of ordinary and exploratory approvals are added to the next visible set, while the remaining

applicants leave the replay. Outcomes are revealed within the simulation round, omitting actual repayment delays.

Write a_i for exposure and set $\mathrm{LGD}_i = 1$. The planning cost, exploration budget, and selected risk are

$$\widehat{c}_{it}^{(m)} = \widehat{p}_{it}^{(m)} a_i, \qquad B_t^{(m)} = \rho \sum_{i \in \mathcal{C}_t^{(m)}} \widehat{c}_{it}^{(m)}, \qquad \widehat{R}_t^{(m)} = \sum_{i \in \mathcal{S}_t^{(m)}} \widehat{c}_{it}^{(m)}. \tag{2}$$

The constraint $\widehat{R}_t^{(m)} \le B_t^{(m)}$ applies to exploratory approvals. Defaulted exposure is evaluated separately as $R_t^{\mathrm{real},(m)} = \sum_{i \in \mathcal{S}_t^{(m)}} y_i a_i$. With unit LGD, this experimental loss proxy omits recovery adjustments and the broader components of IFRS 9 accounting.

Policies share the initial split and exogenous batches, but their feedback choices affect subsequent training sets, predictions, candidates and budgets. Pairing on dataset, seed and ρ compares complete policies under common incoming data. Same-state audits introduced below hold the later acquisition state fixed when the analysis requires a selector-level comparison. Superscripts m are suppressed below for within-policy optimization.

3.2 Lexicographic quantity and composition

LRBE first determines the maximum feasible exploratory count:

$$K_t^* = \max_{z \in \{0,1\}^{|\mathcal{C}_t|}} \sum_i z_i \quad \text{subject to} \quad \sum_i z_i \widehat{c}_{it} \le B_t. \tag{3}$$

For nonnegative costs and unit count value, sorting costs and taking the longest feasible prefix solves this stage in $O(n \log n)$ time. This cardinality construction defines the quantity reference. A fixed, outcome-blind secondary score q_i then determines the composition:

$$\max_z \sum_i z_i q_i \quad \text{subject to} \quad \sum_i z_i = K_t^*, \quad \sum_i z_i \widehat{c}_{it} \le B_t, \quad z_i \in \{0,1\}. \tag{4}$$

The equality prevents the secondary objective from sacrificing feasible count *at the current policy state*. Distinct policies may still acquire different cumulative counts after their histories diverge. Both stages use currently available features and predictions; candidate outcomes and test labels remain unavailable during selection.

3.3 Acquisition policies

CountOnly-MinRisk minimizes predicted risk among maximum-count sets by setting $q_i = -\widehat{c}_i$. It is abbreviated MinRisk in tables. **LRBE-Uncertainty** uses $q_i = \widehat{p}_i(1 - \widehat{p}_i)$. **LRBE-FAVE** uses the frozen FAVE-v2.1 score, a feature-space local information proxy (Appendix A). Its Fisher-style geometry is a surrogate constructed from predictions and covariates, not the Hessian of the fitted boosted-tree model or a guarantee of retraining improvement.

Table 1. Acquisition objects used in the comparison.

Policy	Keeps K_t^*	Objective	Comparison role
MinRisk	Yes	Minimum predicted risk	Risk reference
LRBE-IG/FAVE	Yes	Maximum score	Composition at maximum count
ScoreFirst-IG/FAVE	No	Maximum score	Quantity opportunity cost
Random controls	Varies	Random order or score	Selection-rule control

LRBE-IG uses empirical committee mutual information. At each round, 20 models are fitted to same-size bootstrap samples of the current visible set. If $p_i^{(r)}$ is member r's default probability and $H(p) = -p\log p - (1-p)\log(1-p)$, then

$$q_i^{\mathrm{IG}} = H\left(\frac{1}{20}\sum_{r=1}^{20} p_i^{(r)}\right) - \frac{1}{20}\sum_{r=1}^{20} H\left(p_i^{(r)}\right). \tag{5}$$

Committee members use the canonical XGBoost hyperparameters and visible labels. The committee determines acquisition scores, while the single canonical model defines candidates, predicted costs, and final evaluation. We sum applicant-level scores across the selected set. **ScoreFirst-IG** maximizes the same nonnegative score under the risk constraint without the count equality. This committee score tests the quantity constraint independently of FAVE's approximation.

ScoreFirst policies maximize the corresponding nonnegative uncertainty or FAVE utility under the risk constraint without the count equality. **Joint-Best-Val-Uncertainty** maximizes $\sum_i z_i(\bar{q}_i - \lambda\bar{c}_i)$ over positive-score candidates, where bars denote within-pool min–max normalization. Its parameter is chosen using only the initial visible set, split into 60% training, 20% pseudo-candidates and 20% validation. Validation PR-AUC determines the choice, with Recall and then smaller λ as tie-breakers. These comparators vary how the available risk budget is allocated between cost and acquisition score.

Two additional controls vary randomization. **RiskBounded-Random** visits candidates in a reproducible pseudo-random order and accepts each if its cost fits the remaining budget. **RiskBounded-Epsilon** preserves K_t^* and uses $q_i = -0.9\bar{c}_i + 0.1u_i$, where u_i is a deterministic hash-based random score. We call it the *random-score* control in prose; the value 0.1 weights the random score in the objective. **NoExplore** reveals ordinary approvals only. **Oracle** reveals every candidate outcome and ignores the exploration budget. It serves as a full-feedback diagnostic, although finite-sample retraining can give it a lower test score than a budgeted policy.

4 Experimental Design

4.1 Data and replay protocol

For each seed, a random permutation assigns 25% of records to a fixed outer test set. The first 20% of the remaining stream form the initial visible set, with

Table 2. Exact input cohorts and exposure semantics. Risk units are dataset-specific.

Dataset	Records	Exposure and evidence role
HMEQ	5,960	LOAN amount; development
LendingClub	16,566	loan-amount proxy; external validation
UCI	30,000	unit exposure; development
PPDai	55,596	unit exposure; exploratory external

a minimum of 100 records; all remaining records are split into ten incoming batches. The initial visible set is a random cohort sample. Policy-dependent selective observation begins when policies process the ten subsequent batches: ordinary approvals are observed automatically, whereas exploratory approvals are chosen from the threshold-defined candidate set. We refer to this design as *random-order offline replay*, which represents randomized chronology and simulated approvals.

Fresh confirmation seeds 1001–1010 are disjoint from development seeds 0–19. The fresh design includes an objective comparison at $\rho = 0.1$, a four-policy sweep over $\rho \in \{0.025, 0.05, 0.1, 0.2, 0.4\}$, and NoExplore/Oracle references. The two randomization controls use the same fresh seeds at $\rho = 0.1$. A separate primary IG block at $\rho = 0.1$ reruns MinRisk, LRBE-FAVE, ScoreFirst-FAVE, LRBE-IG, and ScoreFirst-IG together, yielding 200 dynamic episodes from a common evidence block. Development analyses remain separate because their protocols differ from the fresh uncertainty estimates. All reported analyses use the registered seeds, policies, and archived raw outputs.

A controlled one-step experiment uses synthetic tasks with prespecified seeds 2001–2020. Each task contains a selectively observed initial set, one incoming batch, and separate test strata for an initially underrepresented region and its complement. It crosses unit exposure with normalized lognormal exposure and evaluates region-1 weights of 0.1 and 0.5. MinRisk, LRBE-IG, ScoreFirst-IG, and a maximum-count random-score selector share the fitted pre-acquisition state, candidates, costs, numerical budget, and branch-training seed. The experiment measures immediate count sacrifice and predicted-risk use, followed by one retraining step and the specified condition interactions.

4.2 Model, optimization and observation controls

All fresh policies use XGBoost with 100 trees, maximum depth 3, learning rate 0.1, binary logistic objective, and histogram-based training. Numeric features receive median imputation and standardization; categorical features receive a missing-value category and one-hot encoding. Target, ID and the designated exposure column are excluded from predictors. Preprocessing is fitted on visible records only. Each round fits a pre-acquisition model for selection and a post-acquisition model for test evaluation. The outer test set never selects parameters or candidates; population default rates and selected outcomes are used only in offline diagnostics.

Maximum-count selection uses a sorted-cost prefix with a 10^{-10} numerical allowance. The common second-stage binary program and quality-first/joint programs use SciPy's mixed-integer solver; an unsuccessful solve raises an error. Risk auditing permits a 10^{-6} absolute allowance. Candidate costs are strictly positive under the validated exposures and rejection threshold, so zero budgets yield zero feasible count. Empty candidate sets also yield no exploratory selections. GPU training and supported FAVE matrix operations accelerate computation, while optimization remains CPU-side. The registered bootstrap committee runs on CPU with four concurrent workers; the canonical model remains on GPU. Appendix A records implementation and validation details.

4.3 Outcomes and statistical comparisons

We report cumulative exploratory count, cumulative predicted risk, final-test default Recall, and Brier score. Recall is $TP/(TP + FN)$ at $\tau = 0.5$; Brier is the mean squared error of default probabilities. The acquired default fraction describes the selected outcomes. A separate feedback diagnostic, F, is the default fraction in the entire visible set divided by the default fraction in the non-test stream, including ordinary approvals and the initial set. Applicant overlap is measured separately by selected identifiers in the same-state audit.

At each dataset and budget ratio, final policy differences are paired over seeds. Tables show mean differences and pointwise 95% percentile bootstrap intervals from 10,000 seed resamples. We describe close counts empirically; equivalence or noninferiority would require dedicated tests. The comparisons span multiple metrics and conditions, so isolated interval exclusions remain pointwise findings.

The IG block adds a same-state mechanism audit. At each round of the MinRisk trajectory, all five selectors receive the same visible set, candidates, cost vector, and numerical budget. We record count sacrifice $D(S) = K_t^* - |S|$, composition premium $P(S) = \widehat{R}(S) - M(|S|)$, where $M(k)$ is the minimum predicted risk of a size-k set, maximum-count slack, selected IDs, and Jaccard overlap with MinRisk. The identity $B - \widehat{R}(S) = [B - M(K_t^*)] + [M(K_t^*) - M(|S|)] - P(S)$ separates unused budget into maximum-count slack, the minimum risk released by selecting fewer applicants, and the risk premium of the selected composition. Alternative selections are evaluated at the anchor state and do not update its subsequent trajectory, so this audit measures immediate selector behavior.

The controlled experiment applies the same 10,000-draw paired seed bootstrap to LRBE-IG minus ScoreFirst-IG count, LRBE-IG minus MinRisk predicted risk and Brier score, and the prespecified exposure-by-region-weight interactions. Its confirmation run holds training randomness fixed across branches. Candidate-label perturbations leave predictions, candidates, IG scores, and selections unchanged; test-label perturbations leave the fitted predictor unchanged.

For feedback–prediction associations, the primary variables are raw gains from the same-seed NoExplore reference: $\Delta F = F_m - F_N$, $\Delta\text{Recall} = \text{Recall}_m - \text{Recall}_N$, and $\Delta\text{Brier} = \text{Brier}_N - \text{Brier}_m$. Positive Brier gain means improvement. Spearman associations are computed separately within dataset and budget, retaining all methods belonging to a resampled seed in each of 10,000 cluster-

bootstrap draws. We report budget-centered associations as secondary descriptions. Reference-gap ratios appear in the supplement when the oriented Oracle gap is positive; nonpositive or numerically unresolved gaps are N/A. The primary raw-gain analysis avoids division by these unstable reference differences.

Cohort provenance. UCI is the Default of Credit Card Clients dataset [24]; HMEQ is the home-equity cohort documented in [3]. LendingClub merges the documented DePaul-hosted 2014 and 2015 public samples into a mature Fully Paid/Charged Off cohort [7]. Its outcomes concern historically issued loans, not observed outcomes for genuinely rejected applicants. PPDai is linked to Figshare file 26049578 [9]; the current reconstructed source is recorded by hash, but byte parity with the historical processed file is not established. It therefore carries exploratory evidentiary weight and is not pooled with historical PPDai results. Its exposure column is identically one. Hash-checked source instructions identify the exact inputs without redistributing third-party data. The strict exposure audit finds no missing, nonnumeric, nonfinite or nonpositive exposure in the three stored exposure columns; UCI constructs unit exposure directly.

5 Results

5.1 How much quantity does score-first selection sacrifice?

The first question is how many feasible labels a score-first policy gives up. Removing the count equality quantifies that opportunity cost. Relative to ScoreFirst-FAVE, LRBE-FAVE acquires 67.1 [62.6, 71.4] additional labels on HMEQ, 209.6 [202.5, 216.1] on LendingClub, 2.0 [0.9, 3.1] on PPDai, and 56.4 [54.7, 58.2] on UCI. Recall favors LRBE-FAVE on HMEQ by 0.0175 [0.0041, 0.0297]; the other three Recall intervals contain zero. The count constraint guarantees maximum cardinality within each policy's current state. The cumulative differences reported here are empirical outcomes of the resulting policy trajectories, and score-first selection does not produce a consistent predictive improvement across datasets.

The random-order control provides a second view. Against MinRisk, it acquires 87.0 fewer labels on HMEQ, 317.6 fewer on LendingClub, and 59.5 fewer on UCI; PPDai differs by 0.3. HMEQ Recall is lower by 0.0268 [-0.0395, -0.0140], while the other Recall intervals contain zero. The fixed-count random-score control remains within 2.7 labels of MinRisk in every dataset, and all Recall and Brier intervals contain zero. These are complete-policy comparisons over policy-specific trajectories.

FAVE and MinRisk have close observed cumulative counts, an empirical similarity across their policy-specific trajectories. FAVE's count differences are 1.1 [-0.9, 3.0] on HMEQ, -0.1 [-2.5, 2.6] on LendingClub, 0.0 [-1.0, 1.1] on PPDai, and -0.3 [-2.0, 1.5] on UCI. It uses more predicted risk in all four datasets, while every Recall interval contains zero. Three Brier intervals contain zero; UCI favors FAVE by -0.00018 [-0.00028, -0.00008]. The UCI estimate is a dataset-specific benefit amid otherwise uncertain Brier differences.

5.2 Independent information gain separates quantity from score choice

The IG extension tests the quantity constraint with a score independent of FAVE's linearization. All five policies are rerun together at $\rho = 0.1$. Panel A of Table 3 shows sizeable feedback increases for LRBE-IG on HMEQ, LendingClub, and UCI. Recall estimates span both directions, while UCI Brier error increases by 0.000156 [0.000019, 0.000298]. PPDai remains a small-feedback exploratory case.

Table 3. Primary IG contrasts at $\rho = 0.1$ ($n = 10$ paired seeds). Entries are mean left-minus-right differences with pointwise 95% intervals. Positive ΔBrier denotes higher error.

Dataset	ΔK	$\Delta \widehat{R}$	ΔRecall	ΔBrier
Panel A: LRBE-IG − ScoreFirst-IG				
HMEQ	31.3 [28.3,34.5]	5,267.5 [−6,697.3,16,208.6]	.00535 [−.00068,.01170]	−.000302 [−.000937,.000364]
LendingClub	49.5 [46.0,53.1]	−9,907.0 [−21,199.7,−1,269.3]	−.00073 [−.00524,.00425]	−.000070 [−.000251,.000103]
UCI	40.1 [36.8,43.6]	1.624 [.241,2.869]	.00166 [−.00187,.00517]	.000156 [.000019,.000298]
PPDai†	0.2 [0.0,0.5]	.050 [.000,.132]	−.00056 [−.00139,.00000]	−.000029 [−.000066,.000000]
Panel B: LRBE-IG − CountOnly-MinRisk				
HMEQ	0.2 [−1.6,1.7]	28,765.5 [19,361.2,39,425.1]	−.00518 [−.01423,.00388]	−.000068 [−.000479,.000292]
LendingClub	−2.9 [−6.0,0.1]	27,025.7 [19,238.3,35,010.3]	.00045 [−.00286,.00488]	−.000029 [−.000174,.000120]
UCI	−1.2 [−3.0,0.5]	1.827 [1.004,2.733]	−.00025 [−.00270,.00222]	−.000098 [−.000262,.000059]
PPDai†	0.4 [−0.6,1.5]	1.024 [.497,1.606]	−.00052 [−.00270,.00142]	−.000064 [−.000134,.000005]

†The registered reconstructed PPDai artifact provides unit-exposure exploratory evidence and is analyzed separately from the historical PPDai block. Risk units are dataset-specific.

Panel B shows that LRBE-IG and MinRisk have close cumulative counts, while LRBE-IG spends more predicted risk on every dataset. The Recall and Brier estimates span improvement and deterioration. In this block, IG changes composition and risk use more clearly than it changes downstream prediction.

The same-state audit fixes the MinRisk visible set, candidates, costs, and numerical budget. Across 100 states per dataset, LRBE-IG has zero count sacrifice. ScoreFirst-IG gives up 3.46, 5.37, 4.04, and 0.02 applicants per state on HMEQ, LendingClub, UCI, and PPDai, respectively. Its mean Jaccard overlap with MinRisk is 0.436, 0.602, 0.135, and 0.224, confirming that the score changes both cardinality and composition at the anchor state. Alternative sets do not generate later audit states, so the audit ends at the selection outcome.

5.3 Controlled conditions reproduce the selection trade-off

The one-step experiment holds the acquisition state and training randomness fixed while varying exposure heterogeneity and the evaluation weight of an underrepresented region. LRBE-IG selects 2.5 [1.45, 3.65] more applicants than ScoreFirst-IG under unit exposure and 3.2 [1.75, 4.85] more under heterogeneous exposure. Relative to MinRisk, it uses 0.221 [0.165, 0.284] and 0.136 [0.101, 0.171] more predicted-risk units, respectively. All four primary Brier intervals span zero.

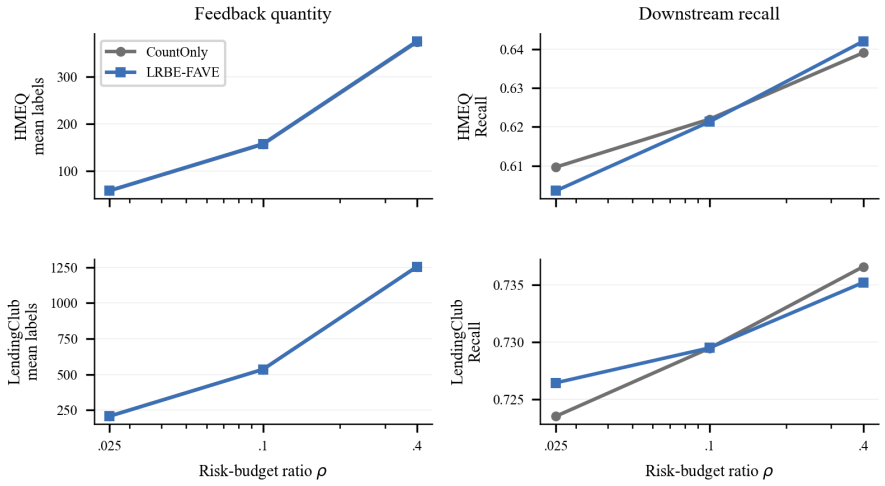


Fig. 1. Mean cumulative exploratory count and final Recall across budget ratios for HMEQ and LendingClub (ten fresh seeds). Larger budgets increase feedback; the Recall ordering of MinRisk and FAVE varies across budgets.

The prespecified condition interactions are also uncertain. Increasing the target-region weight from 0.1 to 0.5 changes the IG-minus-MinRisk Brier contrast by 0.000126 [−0.000005, 0.000255] under unit exposure and −0.000058 [−0.000217, 0.000102] under heterogeneous exposure; the difference between those changes is −0.000184 [−0.000413, 0.000029]. These estimates leave the moderating roles of exposure and target weight unresolved. The incoming batch averages about 3,751 ordinary approvals, compared with exploratory K^* means of 29.55 and 118.7, so the evaluated retraining intervention is small relative to the shared ordinary feedback.

5.4 Budget sensitivity

The budget ratio has a much larger and more consistent relation with acquired quantity than the secondary score. HMEQ MinRisk grows from 57.4 labels at $\rho = 0.025$ to 374.0 at $\rho = 0.4$; FAVE grows from 58.0 to 375.8. LendingClub MinRisk grows from 207.6 to 1,253.1 and FAVE from 206.3 to 1,253.3. Recall moves less and method ordering changes across budgets. Figure 1 shows representative HMEQ and LendingClub trajectories. The sweep changes both the scale of feedback and the subsequent policy state, so it is a sensitivity analysis rather than a quantity-only intervention.

5.5 Feedback diagnostics vary across datasets and outcomes

Normalized recovery divides policy changes by an Oracle–NoExplore gap and represents recovery when that oriented gap is positive. In the present data, the

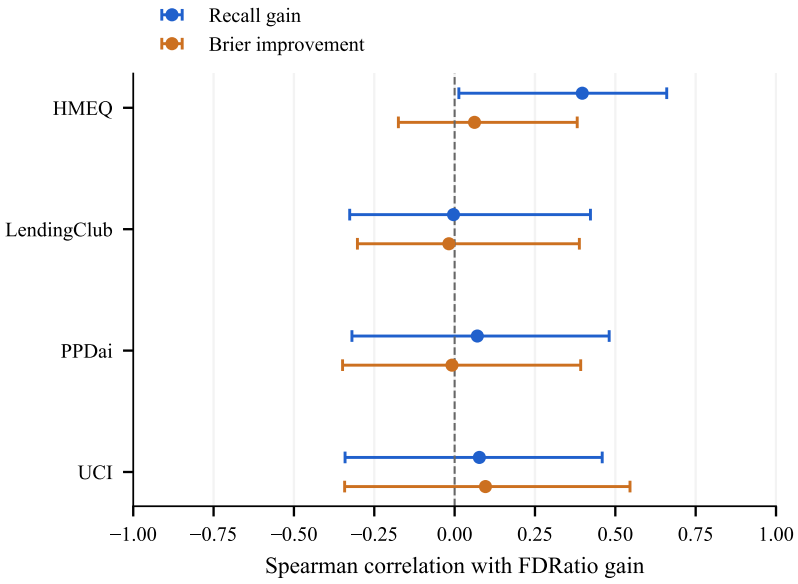


Fig. 2. Within-dataset associations between visible-set default-ratio gain and downstream improvement at $\rho = 0.1$. Points use 40 terminal policy cells (four policies, ten seeds); bars are pointwise 95% intervals from 10,000 whole-seed bootstrap resamples. Positive Brier gain means lower error.

Recall gap is nonpositive in 10 of 40 dataset–seed references: two LendingClub seeds, two PPDai seeds, including one zero, and six UCI seeds. Oracle gaps are also nonpositive in 21 of 40 ROC-AUC and 15 of 40 PR-AUC references. We therefore use raw gains from NoExplore as the primary analysis and reserve normalized recovery for positive-gap cells.

Figure 2 relates visible-set default-ratio gain to Recall and Brier improvement at $\rho = 0.1$. HMEQ has a positive Recall association, $\rho_s = 0.397$ [0.013, 0.660], but its Brier association is uncertain, 0.062 [−0.175, 0.381]. LendingClub, PPDai, and UCI Recall intervals contain zero at this budget, as do all four Brier intervals. Across the five within-budget analyses, HMEQ Recall estimates range from 0.389 to 0.514 and all intervals exclude zero. LendingClub instead has negative Recall associations at $\rho = 0.025$ and 0.05, followed by uncertain estimates. PPDai's smallest-budget cells have near-degenerate ranks and yield perfect boundary coefficients.

These associations concern a diagnostic of the complete visible set, which includes the initial set and ordinary approvals. A larger ratio can move toward or away from the population target, and policy-specific predictors make selected mean PD an unsuitable cross-policy identity measure. Feedback diagnostics align

with Recall in some regimes and diverge in others. We report the correlations as descriptive relationships between complete-policy outcomes.

5.6 Predicted feasibility and realized default exposure

All non-Oracle selections obey the fitted point-cost constraint within audit tolerance. Across 104 equally sized method–dataset–budget groups, the mean rate at which realized defaulted exposure exceeds the policy's numerical predicted budget is 0.4798, with group rates ranging from 0.01 to 0.84. Because each group contains 100 seed–round observations, the unweighted group mean also equals the observation-weighted mean for these included cells. The range summarizes observed groups rather than sampling uncertainty.

This comparison evaluates an ex-post loss exceedance against a budget constructed from fitted expected costs rather than a realized-loss quantile. The exceedance rate shows the gap between point-estimate feasibility and ex-post loss-cap compliance, combining model calibration, stochastic default variation, exposure assumptions, and dependence among loans.

Descriptive computation time. Recorded episode wall time includes training, selection, evaluation, and system effects. Mean LendingClub times in the IG block are 44.0 s for MinRisk, 82.8 s for LRBE-FAVE, and 498.6 s for LRBE-IG; these are end-to-end episode descriptions rather than isolated acquisition latency.

6 Discussion

What a quantity-preserving reference contributes. Maximizing feasible count makes the opportunity cost of a secondary acquisition score observable. Removing the count equality with FAVE, IG, or random-order selection reduces feedback substantially on the three datasets with informative count ranges. The resulting predictive estimates vary by dataset: HMEQ Recall favors LRBE-FAVE, UCI Brier favors ScoreFirst-IG, and the remaining primary intervals span both directions. More feedback is therefore an intermediate outcome whose value must be checked against prediction. Within the eligibility rule and predicted-risk constraint, MinRisk states the least predicted risk required to retain the current maximum count.

Policy effects and composition evidence. The three experimental views answer different questions. Dynamic replay compares complete acquisition policies that begin with the same data stream and then evolve through policy-specific feedback, predictions, candidates, and budgets. Same-state audits fix those quantities and show how selectors exchange count, risk, and score at one decision point. The controlled experiment adds one retraining step after fixing the acquisition state. It reproduces the count and predicted-risk contrasts, while its exposure and target-weight interactions remain uncertain. This division connects long-run policy outcomes to an immediate selection mechanism without treating the two analyses as interchangeable.

Why predictive differences can remain small. Most labels added in the controlled task come from ordinary approvals shared across selectors: about 3,751 per incoming batch, compared with exploratory K^* means of 29.55 under unit exposure and 118.7 under heterogeneous exposure. Differences among exploratory selections therefore modify only a fraction of the post-acquisition training set. This scale contrast offers one plausible account of the small Brier differences. The experiment varies exposure and target weight, leaving dilution as a possible explanation for subsequent study. More broadly, an acquisition score matters to prediction when the labels it changes affect the fitted model and target metric; score totals and feedback composition describe earlier links in that chain.

Scope of inference. The evidence concerns random-order replay on historical cohorts with observed outcomes. LendingClub contains observed loan outcomes rather than outcomes for actual historical rejects, labels arrive immediately in the simulation, and $\tau = 0.5$ supplies a common experimental operating point. PPDai contributes unit-exposure exploratory evidence; HMEQ and LendingClub supply monetary heterogeneity. The registered IG comparison uses $\rho = 0.1$, while the broader budget sweep covers the original policy set. Published online exploration methods motivate the research question, and the randomized controls isolate design choices within the present replay. The conclusions concern these acquisition policies and replay conditions.

Risk interpretation. An optimizer can enforce a sum of predicted losses while realized loss remains variable. Here, an exceedance means that realized defaulted exposure in a seed–round observation is larger than the policy's point-estimate budget. This ex-post comparison with expected loss combines probability error, default randomness, and exposure assumptions. A live credit experiment would additionally specify product-level exposure and recovery, delayed outcomes, concentration and fairness safeguards, and stopping rules. The current point-cost constraint supplies a planning ledger for the replay rather than those operational controls.

7 Conclusion

Risk-budgeted feedback acquisition involves two choices: how many outcomes to make observable and which applicants should provide them. LRBE separates them by preserving the current maximum feasible count before applying a composition score. Across FAVE, committee IG, and randomized controls, removing the count constraint often reduces feedback substantially. Preserving those labels produces dataset-specific predictive changes rather than uniform gains. At similar cumulative counts, FAVE and IG also use more predicted risk than MinRisk, while their Recall and Brier differences remain small and uncertain in the primary comparisons. The controlled analysis reproduces the selection trade-offs, with unresolved exposure and target-weight interactions. Evaluating quantity, predicted-risk use, composition, and downstream performance together

shows where an acquisition score changes the learning process and whether that change reaches the prediction target.

A Secondary Score and Reproducibility

Frozen FAVE-v2.1 score. Let $v_j = [1, x_j^\top]^\top$ be the augmented, preprocessed feature vector. For the current visible set $\mathcal{L}_t$ and label-free reference batch $\mathcal{U}_t$, define

$$H_t = \sum_{j \in \mathcal{L}_t} \max\{\widehat{p}_j(1-\widehat{p}_j), 10^{-6}\} v_j v_j^\top, \tag{6}$$

$$G_t = |\mathcal{U}_t|^{-1} \sum_{j \in \mathcal{U}_t} \widehat{p}_j v_j v_j^\top, \qquad \eta = \max\{10^{-6}\,\mathrm{tr}(H_t)/d, 10^{-10}\}, \tag{7}$$

$$A_t = H_t + \eta I, \qquad \widetilde{w}_i = \max\{\widehat{p}_i(1-\widehat{p}_i)\widehat{p}_i^{0.5}, 10^{-10}\}, \tag{8}$$

$$q_i^{\mathrm{FAVE}} = \frac{\widetilde{w}_i v_i^\top A_t^{-1} G_t A_t^{-1} v_i}{1 + \widetilde{w}_i v_i^\top A_t^{-1} v_i}. \tag{9}$$

Here d includes the intercept coordinate. Predicted probabilities are clipped to $[10^{-9}, 1-10^{-9}]$ before these operations; nonfinite score outputs are zeroed and scores are clipped below at zero. The default-weighting exponents and numerical regularization are fixed. This linearized feature-space construction is used with boosted-tree predictions as a scoring proxy; it is not an exact influence calculation for the tree-fitting procedure. The Joint comparator selects $\lambda \in \{0.01, 0.1, 0.5, 1, 2, 5, 10\}$ using only initial-set validation. The initial split is stratified where possible, with a deterministic random split as fallback.

Input and trajectory audit. Exact-input SHA-256 checks cover the registered HMEQ and LendingClub cohorts. The IG block separately verifies the actual hash of its reconstructed PPDai artifact; that artifact has no claimed byte parity with the historical processed PPDai file. Across the 78,122 records audited in the exposure-hardening block, exposure is numeric, finite, and positive, with PPDai values equal to one; UCI constructs ones directly. The strict input loader rejects invalid exposure instead of replacing it by zero. Eight validation trajectories (four datasets, seed 1001, MinRisk and FAVE) agree with the archived outputs in the checked counts, risk values, and predictive metrics. Of 40 paired post-initial rounds, incoming-batch hashes agree in all 40, while candidates and numerical budgets agree in four, directly documenting policy-specific state evolution. The IG and controlled-condition archives retain their own registered seeds and are not pooled as independent replications. The controlled confirmation uses common branch randomness and label-perturbation checks; all 320 endpoint rows match the original block in selection, risk, Recall, and Brier score. Source hashes, full-precision tables, parameter grids, and executable tests accompany the archive.

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